

We now switch to nonlinear optimization and study

$$(P) \quad \min_{x \in \Omega} f(x) \quad \text{where } \Omega \subset \mathbb{R}^n \text{ (often: } \Omega = \mathbb{R}^n \text{)} \\ f: \Omega \rightarrow \mathbb{R}$$

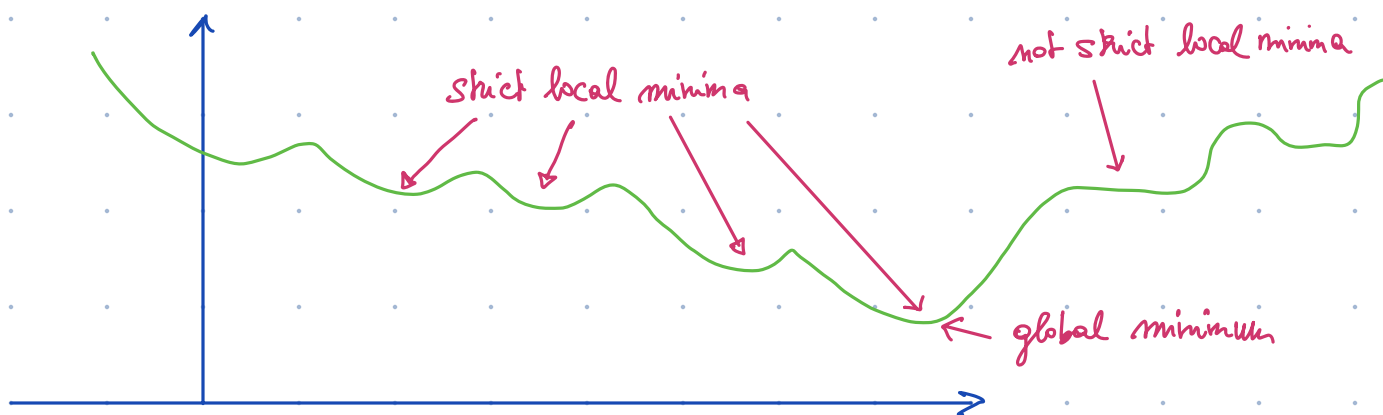
We will answer the question on how to characterize solutions of (P) and how to find them using numerical computation.

Definition: A point $x^* \in \Omega$ is a **local minimum** if there exists an $\varepsilon > 0$ such that for all x with $\|x - x^*\| < \varepsilon$ holds $f(x^*) \leq f(x)$. If this holds with " $<$ " rather than " $\leq$ " for all $x \neq x^*$, it is called **strict local minimum**.

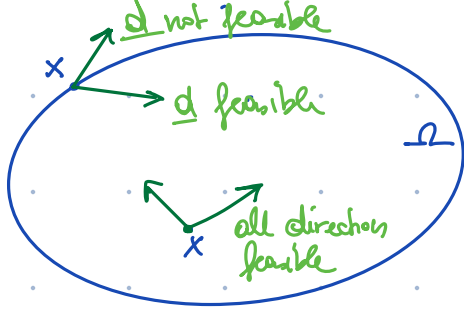
Def: $x^* \in \Omega$ is a **global minimizer** of f over Ω if $f(x^*) \leq f(x)$ for all $x \in \Omega$.

While (P) seeks a global minimizer, we often can only find local minima. Thus, our goal is to find local minima of f .

Sketch of $f: \mathbb{R} \rightarrow \mathbb{R}$:



Def: Given $x \in \Omega$, we say $d \in \mathbb{R}^n$ is a **feasible direction** at x if $\exists \bar{\alpha} > 0$ such that $x + \alpha d \in \Omega$ for all $\alpha \leq \bar{\alpha}$.



Notation and reminders

Assume $f: \Omega \subset \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^1$ (i.e. once continuously differentiable)

$x^* \in \Omega$: $\nabla f(x^*) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x^*) \\ \vdots \\ \frac{\partial f}{\partial x_n}(x^*) \end{bmatrix} \in \mathbb{R}^n$ is the **gradient** of f at x^* .
[typically considered a column vector in this class]

If $f \in C^2$:

$\left[\nabla^2 f(x^*) \right]_{ij} = \frac{\partial^2 f}{\partial x_i \partial x_j}(x^*)$, $\nabla^2 f(x^*) \in \mathbb{R}^{n \times n}$ is the **Hessian** matrix;
 $f \in C^2 \rightarrow$ it is symmetric.

Taylor expansion in 1D: $f: \mathbb{R} \rightarrow \mathbb{R}$, $f \in C^3$:

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2}f''(x_0)(x-x_0)^2 + \underbrace{o(\|x-x_0\|^3)}_{\text{remainder term, goes to zero as fast as } \|x-x_0\|^3}$$

Alternative: (with different remainder term):

$$f(x) = f(x_0) + f'(x_0)(x-x_0) + \frac{1}{2}f''(\tilde{x})(x-x_0)^2 \text{ with } \tilde{x} \in [x_0, x].$$

Taylor expansion in $\mathbb{R}^n$: $f: \mathbb{R}^n \rightarrow \mathbb{R}$, $f \in C^3$:

$$f(y) = f(x) + \nabla f(x)^T (y-x) + \frac{1}{2}(y-x)^T \nabla^2 f(x) (y-x) + o(\|x-y\|^3)$$

$$\square = \square + \boxed{\nabla f(x)^T} \begin{bmatrix} \end{bmatrix} + \begin{bmatrix} \end{bmatrix} \boxed{\nabla^2 f(x)} \begin{bmatrix} \end{bmatrix}$$

Theorem: (First-order necessary condition) Let $\Omega \subset \mathbb{R}^n$, $f \in C^1(\Omega)$.

If $x^* \in \Omega$ is a local minimum of f on Ω , then for any $\underline{d} \in \mathbb{R}^n$ that is feasible at x^* , $\nabla f(x^*)^T \underline{d} \geq 0$.

(Intuition: In every feasible direction, the function f increases!)

Proof: Since $\underline{d}$ feasible at x^* , $x(\alpha) := x^* + \alpha \underline{d} \in \Omega$ for $\alpha \leq \bar{\alpha}$, $\bar{\alpha} > 0$.

Define $g(\alpha) := f(x(\alpha))$.

Then: $g'(\alpha) = \nabla f(x(\alpha))^T \cdot x'(\alpha) = \nabla f(x(\alpha))^T \underline{d}$

$$g(\alpha) = g(0) + g'(0)\alpha + o(\alpha^2)$$

If $g'(0) < 0$, we can find α such that $g(\alpha) < g(0)$, but

this cannot be since $g(0)$ is a local minimum.

Thus: $g'(0) = \nabla f(x^*)^T \underline{d} \geq 0$

□.

Corollary: Let $\Omega = \mathbb{R}^n$. Then if x^* is a local minimum, $\nabla f(x^*) = 0$

Proof: In $\Omega = \mathbb{R}^n$ all directions are feasible

$$\Rightarrow \nabla f(x^*)^T \underline{d} \geq 0 \text{ for all } \underline{d} \Rightarrow \nabla f(x^*) = 0. \quad \square$$

The above also holds in the interior of Ω , where also all directions $\underline{d}$ are feasible.

Example 1: $\min f(x_1, x_2) = x_1^2 - x_1 x_2 + x_2^2 - 3x_2$, $f: \mathbb{R}^2 \rightarrow \mathbb{R}$

$$\nabla f(x) = \begin{bmatrix} \frac{\partial f}{\partial x_1}(x) \\ \frac{\partial f}{\partial x_2}(x) \end{bmatrix} = \begin{bmatrix} 2x_1 - x_2 \\ 2x_2 - x_1 - 3 \end{bmatrix},$$

$$\nabla f(x) = 0 \Leftrightarrow x_1 = 1, x_2 = 2$$

(this is a global minimum, but so far we only know that this is a candidate for a local minimum).

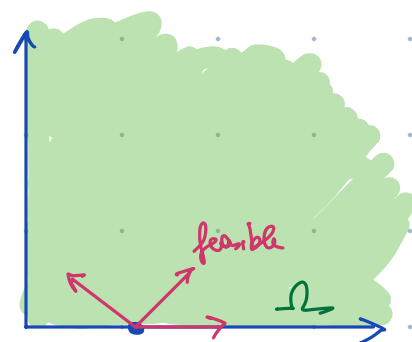
Example 2:

$$\min_{(x_1, x_2) \in \Omega} f(x_1, x_2) = x_1^2 - x_1 + x_2 + x_1 x_2,$$

$$\Omega = \{(x_1, x_2) : x_1 \geq 0, x_2 \geq 0\}$$

Global minimum at $(\frac{1}{2}, 0) =: x^*$

$$\nabla f(x^*) = \begin{bmatrix} 2x_1 - 1 + x_2 \\ 1 + x_1 \end{bmatrix} \bigg|_{x=x^*} = \begin{bmatrix} 0 \\ \frac{3}{2} \end{bmatrix}$$



Feasible directions at x^* are $\{d = \begin{pmatrix} a \\ b \end{pmatrix} : a \in \mathbb{R}, b \geq 0\}$

Thus, for all feasible $\underline{d}$: $\nabla f(x^*)^T \underline{d} = \begin{bmatrix} 0 & \frac{3}{2} \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \frac{3}{2}b \geq 0$
Since $b \geq 0$.

Example 3 (function approximation using least squares)

Given observations $y_i = q_n(x_i)$, $i=1, \dots, m$ for unknown function $q_n: \mathbb{R} \rightarrow \mathbb{R}$

Approximate q by polynomial

$$h(x) = a_n x^n + a_{n-1} x^{n-1} + \dots + a_0, \quad n < m$$

Minimize the square error by choosing coefficients $a_n, a_{n-1}, \dots, a_0$:

$$\mathcal{L}(\underline{a}) = \sum_{i=1}^m (h(x_i) - q(x_i))^2 = \|A \underline{a} - \underline{b}\|^2 = \underline{a}^T A^T A \underline{a} - 2(A \underline{a})^T \underline{b} + \underline{b}^T \underline{b}$$

$$A = \begin{bmatrix} 1 & x_1 & \dots & x_1^n \\ \vdots & \vdots & & \vdots \\ 1 & x_m & \dots & x_m^n \end{bmatrix} \in \mathbb{R}^{m \times (n+1)}, \quad \underline{b} = \begin{bmatrix} q(x_1) \\ \vdots \\ q(x_m) \end{bmatrix}$$

$$\Rightarrow \nabla_{\underline{a}} \mathcal{L}(\underline{a}) = 2A^T A \underline{a} - 2A^T \underline{b} \stackrel{!}{=} 0$$

$$\Rightarrow \underline{A^T A} \underline{a} = A^T \underline{b}$$

$\underline{A^T A} \in \mathbb{R}^{(n+1) \times (n+1)}$

(i.e. solve linear system for $\underline{a}$).